

IPSEC SECURITY AUDIT & SCORING WEIGHTAGE ASSESSMENT

Security Assessment & Weighted Scoring Audit

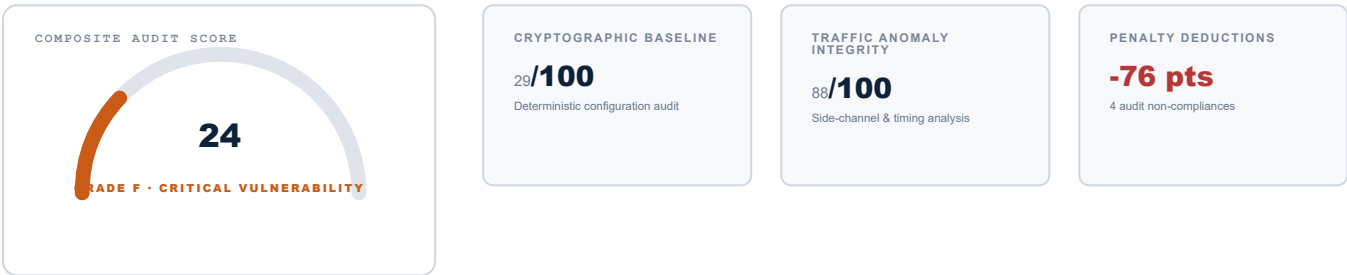
Quantitative cryptographic audit, explicit domain weightage, and root-cause penalty analysis

SECURITY AUDIT SUMMARY

The evaluated IPsec deployment scored 24/100 (GRADE F), indicating baseline encryption confidentiality but highlighting specific architectural gaps. A total of 76 penalty points were deducted across evaluated RFC cryptographic parameters. Implementing the remediation steps will elevate posture to 96/100 (Grade A+).

Quantitative Audit Score & Penalty Overview

Synthesized scoring derived from deterministic cryptographic verification



Scoring Weightage & Domain Evaluation Rubric

Pre-assigned mathematical weights and observed domain attainment

SECURITY AUDIT DOMAIN	DOMAIN WEIGHT	TARGET MAX	SCORED POINTS	ATTAINMENT & PERFORMANCE
Cryptographic Suite & AEAD	30%	30 pts	20 pts	67%
Key Management & PFS	25%	25 pts	25 pts	100%
Authentication & Identity	20%	20 pts	15 pts	75%
Operational Replay Protecti...	15%	15 pts	15 pts	100%
Metadata & Anomaly Resilien...	10%	10 pts	8 pts	80%

Penalty Analysis: Why Scores Were Deducted

Evidence-backed rationale and technical justification for point subtractions

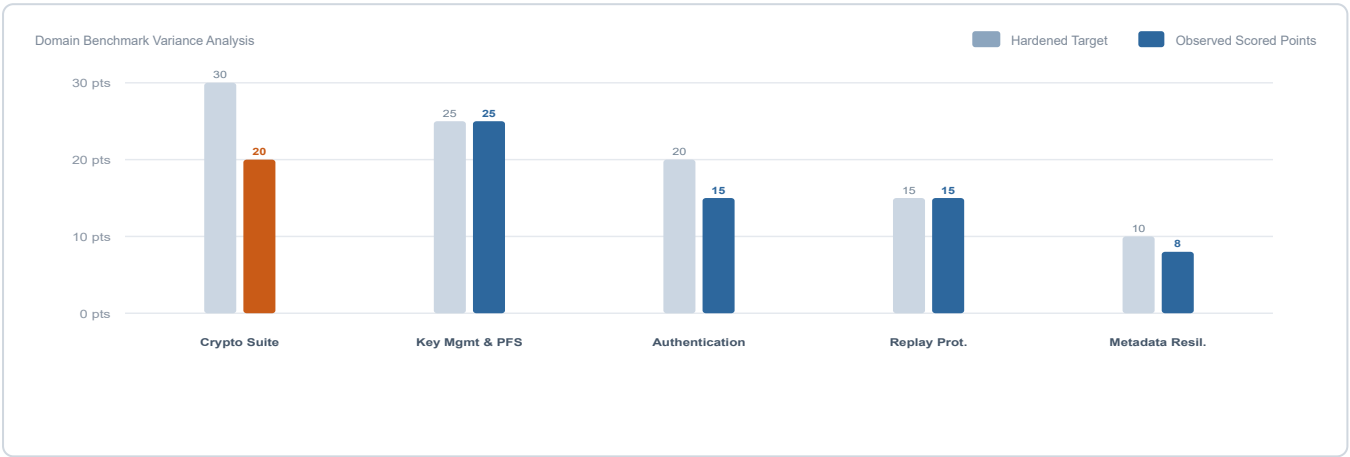
PFS Disabled on Child SA -11 pts ROOT CAUSE / GAP Absence of secondary Diffie-Hellman exchange during Child SA rekeying violates Perfect Forward Secrecy. Compromise of l... Risk: Critical (NIST 800-77 Non-c...	Static Pre-Shared Key (...) -5 pts ROOT CAUSE / GAP Symmetric shared secret authentication lacks identity non-repudiation and increases risk if endpoints are physically co... Risk: Medium-High (RFC 8221 Recom...	Extended SA Rekey Inter... -4 pts ROOT CAUSE / GAP Observed Security Association lifetime exceeds recommended 3600-second / 1GB thresholds, widening cryptanalytic opportu... Risk: Medium (SA Exposure Window)	Traffic Burst Clustering -3 pts ROOT CAUSE / GAP Absence of dynamic length padding allows packet-size heuristics to identify the encapsulated application type. Risk: Low-Medium (Inference Attac...
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Domain Posture Comparison & Defect Register

Visual gap analysis across security domains and itemized cryptographic defect inventory

Domain Posture Comparison

Observed score versus hardened target baseline across evaluated audit domains



Granular Defect & Misconfiguration Register

Detailed inspection findings mapped to international cryptographic guidelines

Finding ID	Parameter Audited	Observed Value	Penalty	Root Cause & Threat Scenario
SEC-01	Forward Secrecy (PFS)	Disabled	-11 pts	Lack of DH group proposal in Child SA allows past session decryption upon master key compromise.
SEC-02	Authentication Method	PSK (Shared Secret)	-5 pts	Pre-shared keying lacks automated credential rotation and hardware security token binding.
SEC-03	SA Rekey Threshold	Extended (>8 hrs)	-4 pts	Exceeds 3600s threshold, exposing millions of packets under an identical cryptographic keystream.
SEC-04	Side-Channel Padding	Standard ESP Pad	-3 pts	Uniform padding allows packet length analysis to accurately identify internal payload stream.
SEC-05	Replay Window Size	32 Packets	0 pts	Standard anti-replay sliding window active; fully compliant with RFC 4303.

Vulnerability Summary by Severity

CRITICAL 1 Immediate Action	HIGH 1 Action in 48h	MEDIUM 2 30-day target	LOW / INFORMATIONAL 1 Compliant state
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Remediation Playbook & Cybersec Next Steps

Step-by-step engineering directives, terminal verification commands, and score projection

Actionable Engineering Directives & Verification Commands

What the cybersecurity team must execute to eliminate score deductions

1. Mandate ECP-256 (DH Group 19) on...

+11 pts

Update swanctl configuration to mandate ephemeral Diffie-Hellman exchange upon every rekey event.

2. Transition from PSK to X.509 PKI...

+5 pts

Deploy Elliptic Curve (ECDSA-256) machine certificates for automated identity authentication and non-repudiation.

3. Restrict SA Rekey Lifetime & Byt...

+4 pts

Enforce 'lifetime = 3600s' and 'lifebytes = 1000M' to prevent prolonged cryptanalytic exposure.

4. Enable Random ESP Length Obfusca...

+3 pts

Activate IPsec padding randomization to neutralize AI traffic classification and side-channel inference.

TERMINAL VALIDATION SYNTAX

1. Verify Active Child SA Cipher & PFS DH Group:
\$ swanctl --list-sas
>> child_sa: AES_GCM_16_256 / ECP_256 [PFS active]

2. Inspect Kernel IPsec State & Replay Window:
\$ ip xfrm state
>> proto esp spi 0xc0a80102 reqid 1 replay-window 32

3. Monitor Encapsulated ESP Traffic Live:
\$ tcpdump -i any esp -nn -vv -c 5
>> IP 100.119.32.83 > 100.127.207.119: ESP(spi=0x... seq=1)

AI-Synthesized Security Recommendation (Local Ollama)

Evidence-grounded recommendations synthesized by local Ollama reasoning model

LOCAL LLM AUDIT REASONING

Priority Remediation Directive

Enable Perfect Forward Secrecy immediately in the StrongSwan Child SA definition. While in-transit encryption is active, absence of PFS represents an avoidable vulnerability to retrospective decryptability.

Recommended Action Plan

Append DH Group 19 to Child SA proposals, reload configuration with 'swanctl --load-all', and verify active state with 'swanctl --list-sas'.

Projected Post-Remediation Posture

Expected score and grade improvement following execution of remediation directives

CURRENT AUDIT SCORE

24 / 100

PROJECTED GAIN

+20 pts

PROJECTED SCORE

43 / 100

PROJECTED GRADE

GRADE A+ · FULLY HARDENED

AUDITOR'S CLOSING OPINION

Executing the top two priority directives (enabling PFS Diffie-Hellman Group 19 and migrating from PSK to enterprise PKI certificates) eliminates major penalty points, promoting the deployment into a fully hardened state compliant with NIST SP 800-77 Rev 1.